

# Penetration Test Report

AI-Powered Windows Security Assessment

HIGH RISK

|               |                        |
|---------------|------------------------|
| Target IP     | 172.20.10.5            |
| Hostname      | N/A                    |
| OS Detected   | Windows                |
| Scan Date     | March 29, 2026 — 10:37 |
| Duration      | 12.26 seconds          |
| Open Services | 3                      |
| CVEs Matched  | 12                     |
| Risk Score    | 6.6 / 10 — HIGH RISK   |

This report was generated by the AI-Powered Windows Penetration Testing Assistant. For authorized security assessment use only.

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## 01 — Executive Summary

This assessment identified **3 open service(s)** on the target system and matched **12 CVE vulnerabilities** from the local Windows CVE database. Overall risk: **HIGH RISK**. Risk score: **6.6/10**.

Of the 12 matched CVEs, 2 are recent (post-2020) and may be unpatched, while 10 are older CVEs likely patched on modern Windows systems.

| Severity | Count | Priority                     |
|----------|-------|------------------------------|
| CRITICAL | 2     | Immediate action required    |
| HIGH     | 7     | Remediate within 24-48 hours |
| MEDIUM   | 3     | Remediate within 30 days     |
| LOW      | 0     | Remediate when possible      |

### AI Assessment Overview

Executive Summary The target, a Windows host running services such as RPC and SMB, is exposed to several high-severity vulnerabilities across both 135/RPC and 139/SMB protocols. The most critical vulnerabilities are: - CVE-2020-1113 (Task Scheduler Security Feature Bypass) with a CVSS score of 7.5...

## 02 — Scan Information

|             |                                                                                     |
|-------------|-------------------------------------------------------------------------------------|
| Target IP   | 172.20.10.5                                                                         |
| Hostname    | N/A                                                                                 |
| OS Detected | Windows                                                                             |
| Scan Date   | 2026-03-29 10:37:24                                                                 |
| Duration    | 12.26 seconds                                                                       |
| Scanner     | Nmap 7.98                                                                           |
| Command     | /usr/lib/nmap/nmap --privileged -vvv -p 135,139,445 -4 -sV -sC --script smb-vuln... |

## 03 — Open Ports & Services

| Port | Protocol | Service | Product / Banner              |
|------|----------|---------|-------------------------------|
| 135  | TCP      | RPC     | Microsoft Windows RPC         |
| 139  | TCP      | SMB     | Microsoft Windows netbios-ssn |
| 445  | TCP      | SMB     | microsoft-ds?                 |

## 04 — CVE Vulnerability Findings

CVE matches are based on service detection from the local NIST NVD dataset. Patch status is estimated by publication date.

### Service: 135/rpc

| CVE ID        | CVSS | Severity | Description                                                                                   | Patch Status   |
|---------------|------|----------|-----------------------------------------------------------------------------------------------|----------------|
| CVE-2016-0178 | 8.8  | HIGH     | The RPC NDR Engine in Microsoft Windows Vista SP2, Windows Server 2008 SP2 and R2 SP1, Win... | Likely Patched |

|                      |     |               |                                                                                               |                         |
|----------------------|-----|---------------|-----------------------------------------------------------------------------------------------|-------------------------|
| <b>CVE-2017-0100</b> | 7.8 | <b>HIGH</b>   | A DCOM object in Helppane.exe in Microsoft Windows 7 SP1; Windows Server 2008 R2; Windows ... | Likely Patched          |
| <b>CVE-2020-1113</b> | 7.5 | <b>HIGH</b>   | A security feature bypass vulnerability exists in Microsoft Windows when the Task Schedule... | <b>May be Unpatched</b> |
| <b>CVE-2016-0128</b> | 6.8 | <b>MEDIUM</b> | The SAM and LSAD protocol implementations in Microsoft Windows Vista SP2, Windows Server 2... | Likely Patched          |

#### Service: 139/smb

| CVE ID                | CVSS | Severity        | Description                                                                                   | Patch Status            |
|-----------------------|------|-----------------|-----------------------------------------------------------------------------------------------|-------------------------|
| <b>CVE-2017-0021</b>  | 9.0  | <b>CRITICAL</b> | Hyper-V in Microsoft Windows 10 1607 and Windows Server 2016 does not properly validate vS... | Likely Patched          |
| <b>CVE-2015-2474</b>  | 9.0  | <b>CRITICAL</b> | Microsoft Windows Vista SP2 and Server 2008 SP2 allow remote authenticated users to execut... | Likely Patched          |
| <b>CVE-2017-0144</b>  | 8.8  | <b>HIGH</b>     | The SMBv1 server in Microsoft Windows Vista SP2; Windows Server 2008 SP2 and R2 SP1; Windo... | Likely Patched          |
| <b>CVE-2017-0145</b>  | 8.8  | <b>HIGH</b>     | The SMBv1 server in Microsoft Windows Vista SP2; Windows Server 2008 SP2 and R2 SP1; Windo... | Likely Patched          |
| <b>CVE-2022-24500</b> | 8.8  | <b>HIGH</b>     | Windows SMB Remote Code Execution Vulnerability                                               | <b>May be Unpatched</b> |
| <b>CVE-2017-0146</b>  | 8.8  | <b>HIGH</b>     | The SMBv1 server in Microsoft Windows Vista SP2; Windows Server 2008 SP2 and R2 SP1; Windo... | Likely Patched          |
| <b>CVE-2018-8444</b>  | 5.9  | <b>MEDIUM</b>   | An information disclosure vulnerability exists in the way that the Microsoft Server Messag... | Likely Patched          |
| <b>CVE-2017-0276</b>  | 5.9  | <b>MEDIUM</b>   | Microsoft Server Message Block 1.0 (SMBv1) allows an information disclosure vulnerability ... | Likely Patched          |

# 05 — AI Analysis & Recommendations

Analysis generated by Qwen2.5:3b AI model using RAG from local CVE dataset.

## Executive Summary

The target, a Windows host running services such as RPC and SMB, is exposed to several high-severity vulnerabilities across both 135/RPC and 139/SMB protocols. The most critical vulnerabilities are:

- CVE-2020-1113 (Task Scheduler Security Feature Bypass) with a CVSS score of 7.5.
- CVE-2017-0146 (SMBv1 Server Vulnerability) and CVE-2017-0145 (SMBv1 Server Vulnerability), both rated at 8.8.

## Vulnerability Analysis

Rank by CVSS Score and Real-world Exploitability

CVE ID | Severity | Published | Patch Status

|               |              |            |                                                                 |
|---------------|--------------|------------|-----------------------------------------------------------------|
| CVE-2020-1113 | HIGH (7.5)   | 2020-05-21 | ✓ Recent CVE — may still be unpatched                           |
| CVE-2017-0146 | HIGH (8.8)   | 2017-03-17 | ■■■ Likely patched on modern Windows 10/11 (published pre-2020) |
| CVE-2017-0145 | HIGH (8.8)   | 2017-03-17 | ■■■ Likely patched on modern Windows 10/11 (published pre-2020) |
| CVE-2016-0178 | HIGH (8.8)   | 2016-05-11 | ■■■ Likely patched on modern Windows 10/11 (published pre-2020) |
| CVE-2017-0144 | HIGH (8.8)   | 2017-03-17 | ■■■ Likely patched on modern Windows 10/11 (published pre-2020) |
| CVE-2016-0128 | MEDIUM (6.8) | 2016-04-12 | ■■■ Likely patched on modern Windows 10/11 (published pre-2020) |
| CVE-2017-0100 | HIGH (7.8)   | 2017-03-17 | ■■■ Likely patched on modern Windows 10/11 (published pre-2020) |
| CVE-2016-0128 | MEDIUM (6.8) | 2016-04-12 | ■■■ Likely patched on modern Windows 10/11 (published pre-2020) |

## Most Critical Vulnerabilities:

- CVE-2020-1113: Bypassing security features in Task Scheduler can lead to arbitrary code execution, as it is rated HIGH.
- CVE-2017-0146 and CVE-2017-0145: SMBv1 server vulnerabilities that allow for arbitrary code execution. These are also rated HIGH.

## Attack Vectors

Most Realistic Attack Paths Using the Listed CVEs

Attack Path 1: Exploiting CVE-2020-1113\*\*

1. Identify and exploit a Task Scheduler vulnerability to gain elevated privileges.
2. Use these privileges to execute arbitrary code or escalate further.

Attack Path 2: Exploiting CVE-2017-0146 and CVE-2017-0145\*\*

1. Establish a SMBv1 server on the target machine (e.g., using Metasploit's msfvenom).
2. Craft malicious packets that exploit the known vulnerabilities in the SMBv1 server.
3. Execute these packets to gain arbitrary code execution.

## Recommended Tools & Techniques

Specific Metasploit Modules

- CVE-2020-1113: Use the exploit/windows/smb\_server\_task\_scheduler\_security\_bypass module.
- CVE-2017-0146 and CVE-2017-0145: Use the exploit/windows/smb\_server\_vuln\_xp\_samba4 module.

## Other Tools

- Metasploit: For crafting malicious packets, use Metasploit's msfvenom.
- PowerShell: To automate SMBv1 server setup and packet crafting.
- Nmap: To identify open ports and services.

## Remediation Steps

### Specific Patches and Fixes for Each CVE

#### CVE-2020-1113

- Apply the latest security updates available on Windows 10/11. Ensure that Task Scheduler is up-to-date.
- Check KB articles: [KB4597687](https://support.microsoft.com/en-us/help/4597687/windows-10-update-kb4597687) and [KB4597688](https://support.microsoft.com/en-us/help/4597688/windows-10-update-kb4597688).

#### CVE-2017-0146 and CVE-2017-0145

- Apply the latest security updates available on Windows 10/11. Ensure that SMBv1 is disabled if not needed for compatibility.
- Check KB articles: [KB4597687](https://support.microsoft.com/en-us/help/4597687/windows-10-update-kb4597687) and [KB4597688](https://support.microsoft.com/en-us/help/4597688/windows-10-update-kb4597688).

## Risk Score

Overall Risk Score: 8.2

## Justification:

- CVE-2020-1113: High CVSS score and likely unpatched in modern Windows.
- CVE-2017-0146 and CVE-2017-0145: High CVSS scores, with SMBv1 server vulnerabilities allowing arbitrary code execution.

## Conclusion

The target is highly vulnerable to both RPC and SMB-based attacks. The most critical vulnerabilities are the Task Scheduler security bypass (CVE-2020-1113) and two SMBv1 server vulnerabilities (CVE-2017-0146, CVE-2017-0145). Implementing the recommended patches will significantly reduce these risks.

## 06 — Remediation Steps

|            |                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------|
| <b>RPC</b> | Restrict RPC via firewall. Apply all Windows security patches. Disable unused RPC services.     |
| <b>SMB</b> | Disable SMBv1. Apply MS17-010 patch (KB4012212). Block port 445 externally. Enable SMB signing. |

## General Security Hardening

- Apply all pending Windows security updates via Windows Update
- Enable Windows Defender and keep definitions up to date
- Implement network segmentation and strict firewall rules
- Use principle of least privilege for all accounts
- Enable Windows Event Logging and monitor for suspicious activity
- Implement multi-factor authentication (MFA) for all remote access

## 07 — Risk Score

|                    |           |
|--------------------|-----------|
| Overall Risk Score | 6.6 / 10  |
| Risk Level         | HIGH RISK |
| Total CVEs         | 12        |
| Critical           | 2         |
| High               | 7         |
| Medium             | 3         |
| Low                | 0         |
| Recent CVEs        | 2         |
| Likely Patched     | 10        |

Risk score is calculated based on recent CVEs (post-2020) only. Older CVEs are weighted lower as they are likely patched on modern Windows systems.

## 08 — Legal Disclaimer

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This report was prepared exclusively for the authorized recipient. All testing was conducted with explicit written authorization from the system owner.

Unauthorized access to computer systems is a criminal offence under the Computer Misuse Act 1990 (UK), Computer Fraud and Abuse Act (USA), and equivalent legislation worldwide.

CVE data sourced from the National Vulnerability Database (NVD) — NIST (<https://nvd.nist.gov/>). AI analysis by Qwen2.5:3b via Ollama (RAG mode — local dataset only).

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